A NODAL CURVE OF GENUS AT LEAST 2 WITHOUT RATIONAL TAILS OR BRIDGES HAS NO GLOBAL VECTOR FIELDS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the normalization transport, the three component cases with their fixed-point counts, the exclusion of the glued, bridge, and tail configurations, and the concluding three-point vanishing are all retained verbatim.

Editorial note. The label lemma-stable-vector-fields names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Derivations across the normalization at nodes. Same chapter, Lemma lemma-nodal-vector-fields. Transports a derivation D on X to a derivation D' on the normalization X^nu whose behaviour at the preimages of nodes mirrors D.
- Vanishing of derivations on smooth projective curves. Same chapter, Lemma lemma-smooth-vector-fields. Part (1) kills derivations on components of genus > 1, part (2) on genus-1 components fixing a point, part (3) on rational components fixing three points.
- The excluded configurations. Same chapter, Examples example-rational-tail and example-rational-bridge. Rule out the low-node configurations that would let a rational or elliptic component survive with a nonzero derivation.

Source reference. Original reference 0E68: sources/stacks-curves/pinned-original-source.tex, source lines 6916–6916. Prerequisite pointer invoked at proof line(s) 6972; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0E67: sources/stacks-curves/pinned-original-source.tex, source lines 6878–6878. Prerequisite pointer invoked at proof line(s) 6975; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let k be an algebraically closed field. Let X be an at-worst-nodal, proper, connected 1-dimensional scheme over k. Assume the genus of X is at least 2 and that X has no rational tails or bridges. Then $Der_k(\mathcal{O}_X, \mathcal{O}_X) = 0$.*

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Proof. Let $D \in \mathrm{Der}_k(\mathcal{O}_X, \mathcal{O}_X)$. Let X^ν be the normalization of X. Let $D' \in \mathrm{Der}_k(\mathcal{O}_{X^\nu}, \mathcal{O}_{X^\nu})$ be the element corresponding to D via Lemma 0E68. Let $C \subset X^\nu$ be an irreducible component. If the genus of C is > 1, then $D'|_{\mathcal{O}_C} = 0$ by Lemma 0E67 part (1). If the genus of C is 1, then there is at least one closed point c of C which maps to a node on X (since otherwise $X \cong C$ would have genus 1). By the correspondence this means that $D'|_{\mathcal{O}_C}$ fixes c hence is zero by Lemma 0E67 part (2). Finally, if the genus of C is zero, then there are at least 3 pairwise distinct closed points $c_1, c_2, c_3 \in C$ mapping to nodes in X, since otherwise either X is C with two points glued (two points of C mapping to the same node), or C is a rational bridge (two points mapping to different nodes of X), or C is a rational tail (one point mapping to a node of X). These three possibilities are not permitted since C has genus ≥ 2 and has no rational bridges, or rational tails. Whence $D'|_{\mathcal{O}_C}$ fixes c_1, c_2, c_3 hence is zero by Lemma 0E67 part (3). $\square$